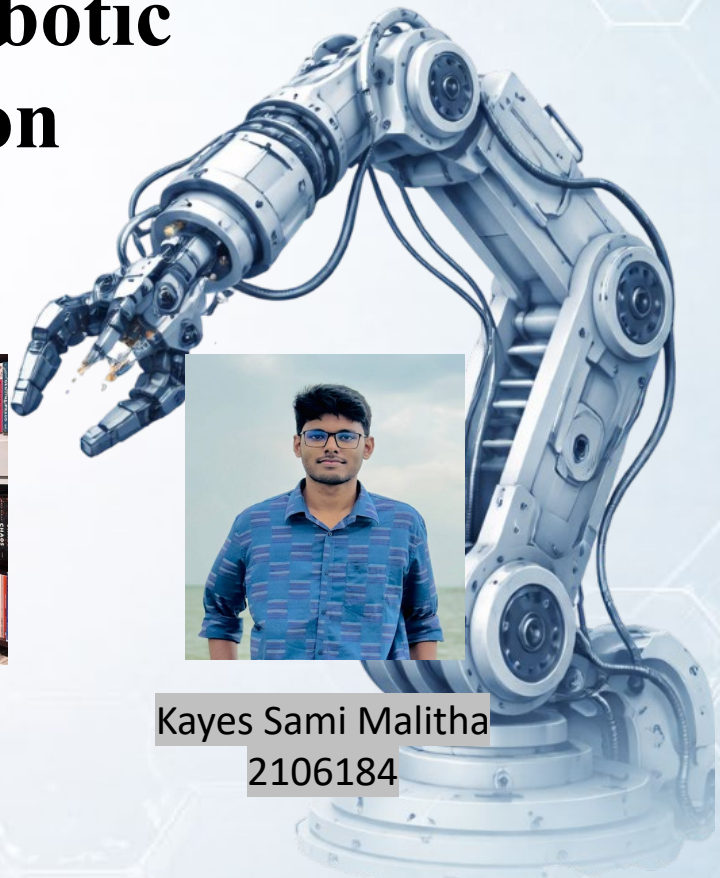
EEE 318 – Control System I Laboratory
Level-3 Term-2 Section: C2
Project Presentation

A Multi-Sensor Controlled 6-DOF Robotic Arm for Human Motion Replication



Submitted By – Group C2.05

Tonmoy Deb
2106177

Mushfiqur Rahman
2106181

Chayan Paul
2106182

Shamin Yeaser
2106183

Kayes Sami Malitha
2106184

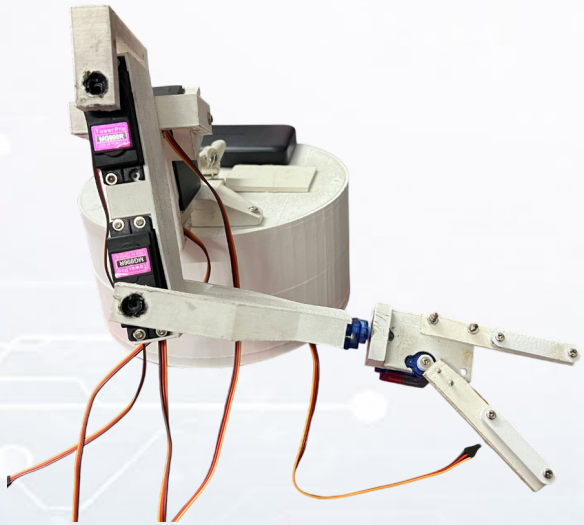
Bangladesh University of Engineering and Technology
Department of Electrical and Electronic Engineering

Outline

- Introduction
- Workflow
- Algorithm Analysis
- Code
- Circuit Diagram
- Cost
- Ethics
- Reference

1. Introduction

This project presents a gesture-controlled robotic arm that can mimic human hand movements. It uses 6 servo motors for motion control, along with 3 accelerometer sensors, a compass sensor and a linear Hall effect sensor to detect orientation and movement. The system captures hand gestures and translates them into corresponding robotic arm actions, enabling tasks like gripping and object handling.



2. Workflow:

Sensor Array
Three gyroscope sensors, one magnetometer, and one Hall-effect sensor are mounted on the hand

↓

I²C Multiplexer Interface
Facilitates communication with multiple sensors over a shared bus

↓

Microcontroller Processing (Arduino UNO)
Central processing of sensor data, smoothing raw data, and control algorithms

↓

Angle Estimation
Computation of roll, pitch, and yaw angles using appropriate mathematical models

↓

PWM Driver Module
Generation of PWM signals to control multiple motors

↓

Command Execution
Transmission of control signals to servo motors

3. Algorithm Analysis

3.1 Shoulder Control

A single gyroscope is mounted on the upper arm to measure pitch motion and lateral (side-to-side) elevation of the arm.

The formulas used are :

$$\text{Shoulder Pitch} = \tan^{-1}\left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}}\right)$$

$$\text{Sideways Elevation} = \tan^{-1}\left(\frac{a_z}{\sqrt{a_x^2 + a_y^2}}\right)$$

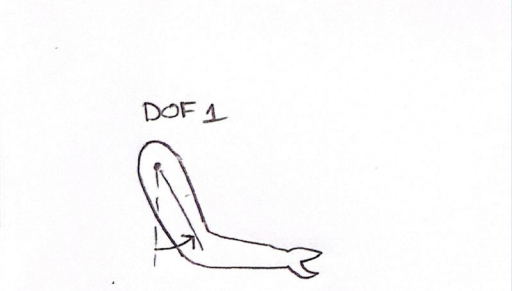
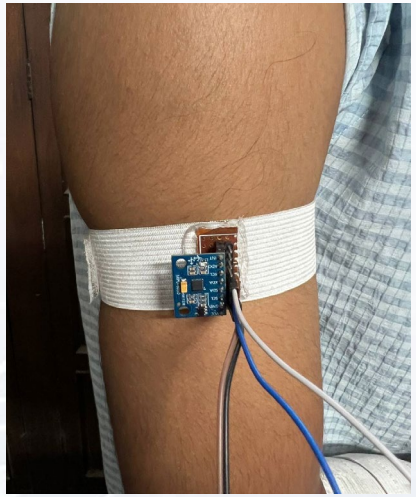
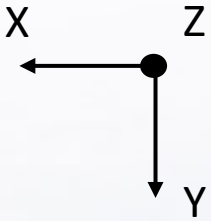


Fig: Shoulder Pitch





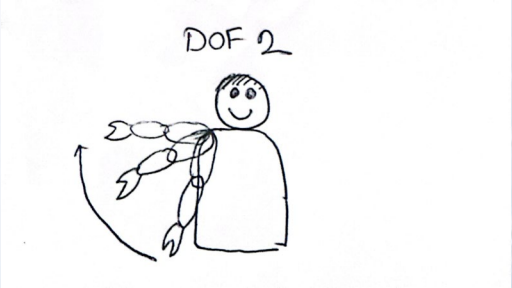


Fig: Sideways Elevation

3. Algorithm Analysis

3.2 Elbow Control

One gyroscope is mounted right after the elbow joint in the lower arm, sideways. Here, we calculated elbow pitch, but since the shoulder angle is taken as the reference and moving the shoulder also moves the elbow, we subtracted shoulder pitch from elbow pitch to find the actual relative angle between the joints.

The formulas used are :

$$\text{Elbow Pitch} = \tan^{-1}\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right)$$

$$Actual\ Elbow\ Angle\ = Elbow\ Pitch\ - Shoulder\ Pitch$$

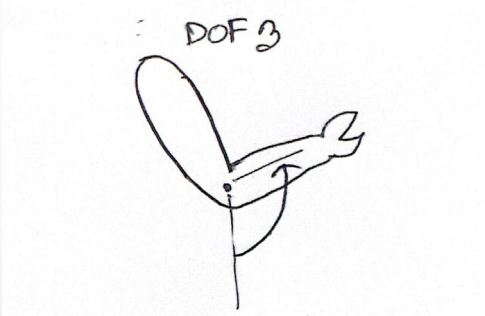
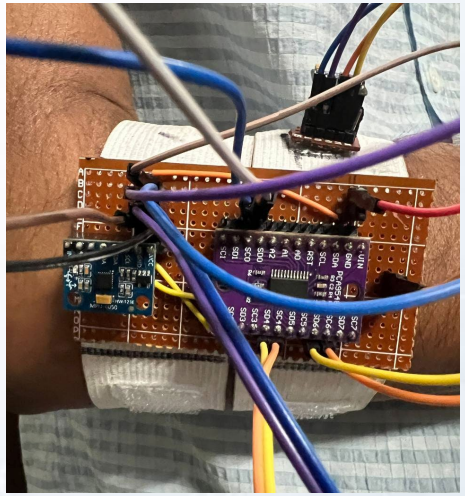
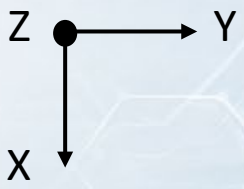


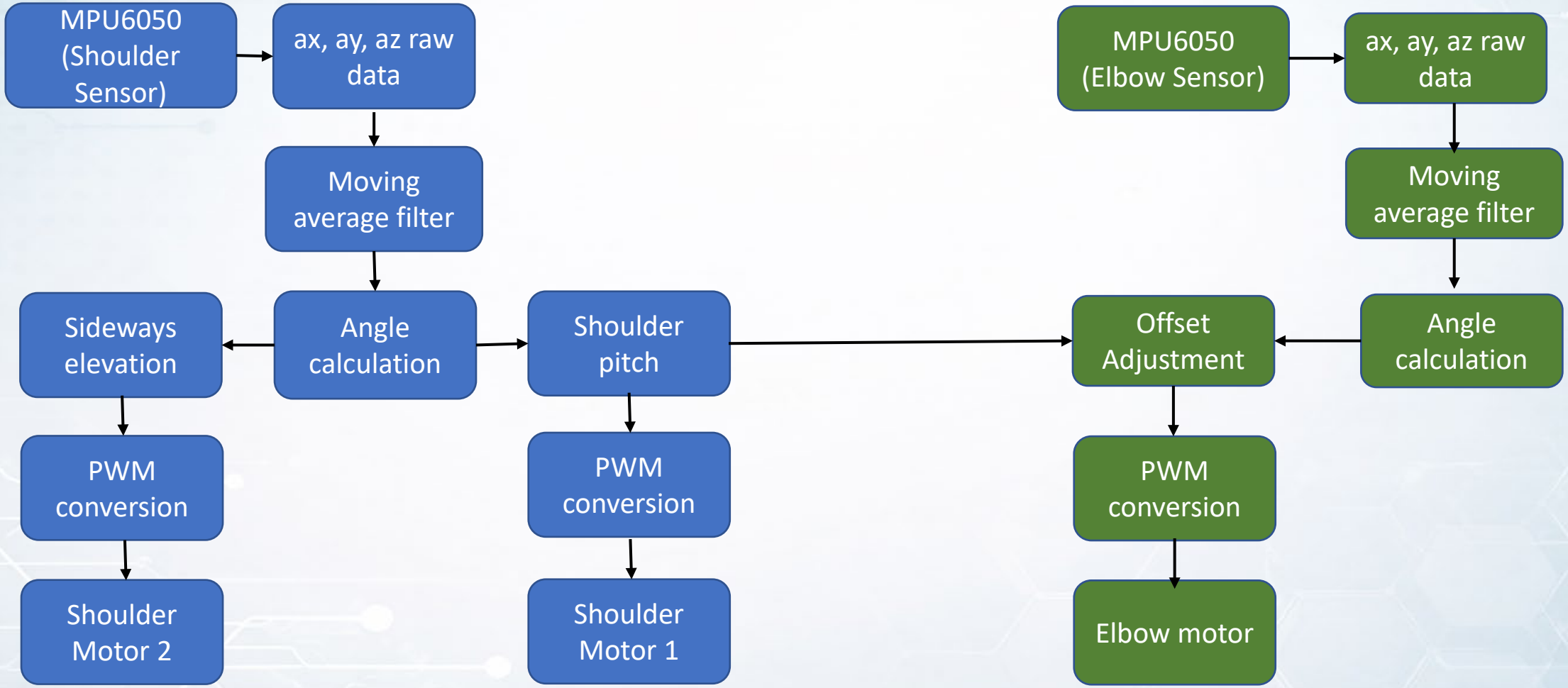
Fig: Elbow Pitch





3. Algorithm Analysis

3.2.1 Shoulder and Elbow Control Flow Chart



50Hz PWM signal is used

3. Algorithm Analysis

3.3 Base Control

A magnetometer is used to control the base rotation of the robotic arm by mapping the hand's horizontal yaw. The initial Earth's magnetic field orientation is set as the reference (90° servo position), and the current azimuth angle is compared to this reference to determine relative left–right orientation.

To ensure the roll and pitch of the hand do not interfere with the horizontal azimuthal angle, we used a compensated formula for azimuth

$X_h = Mx \cos\theta + Mz \sin\theta$

$Y_h = Mx \sin\varphi \sin\theta + My \cos\varphi - Mz \sin\varphi \cos\theta$

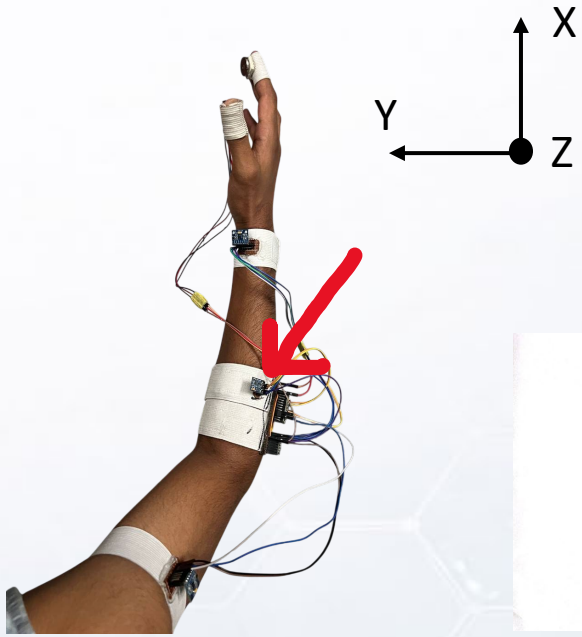
Base Angle $= \tan^{-1} (Y_h / X_h)$

Here,

X_h and Y_h correspond to the horizontal components of the magnetometer data

θ = Elbow Pitch

φ = Roll angle



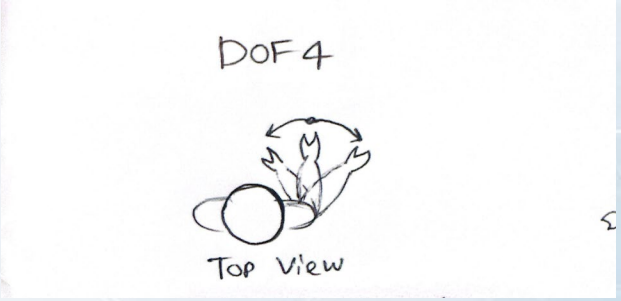
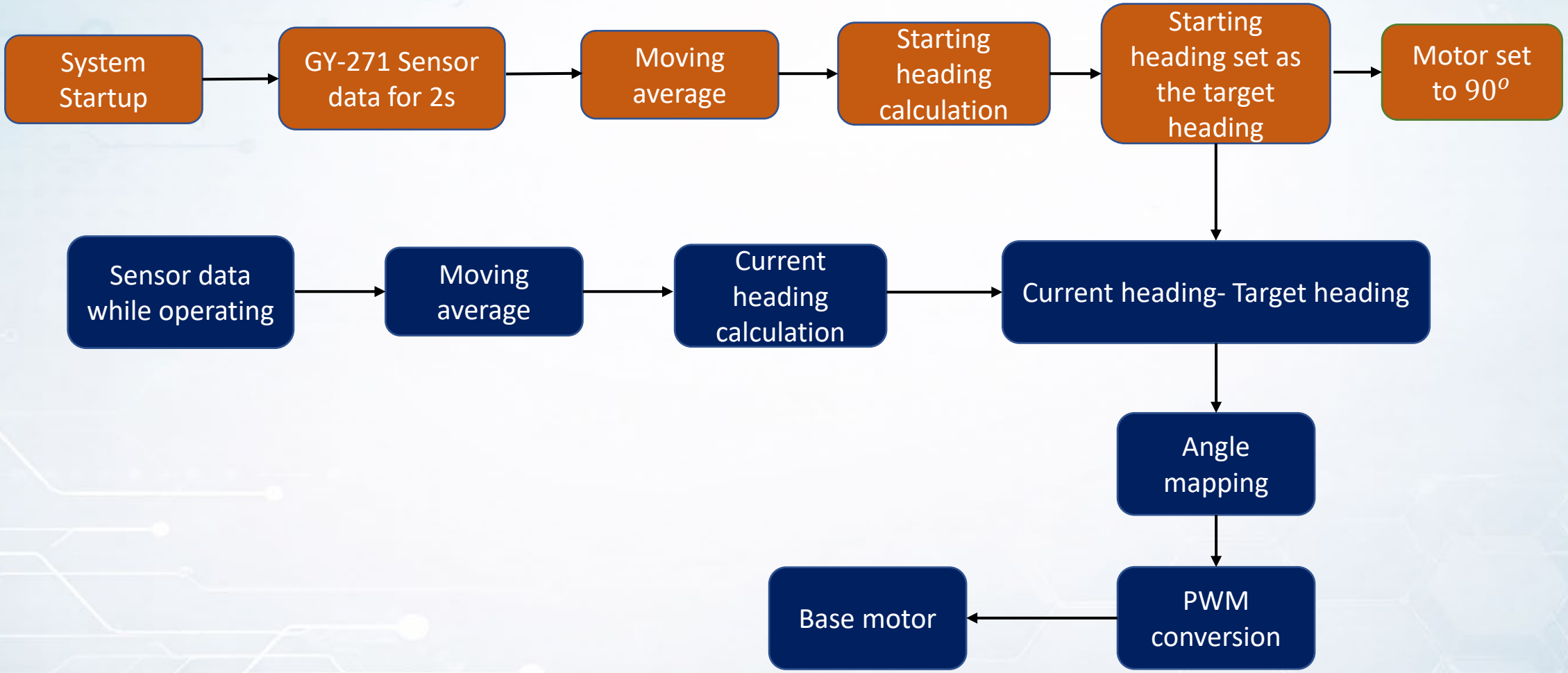


Fig: Yaw Movement

3. Algorithm Analysis

3.3 Base Control

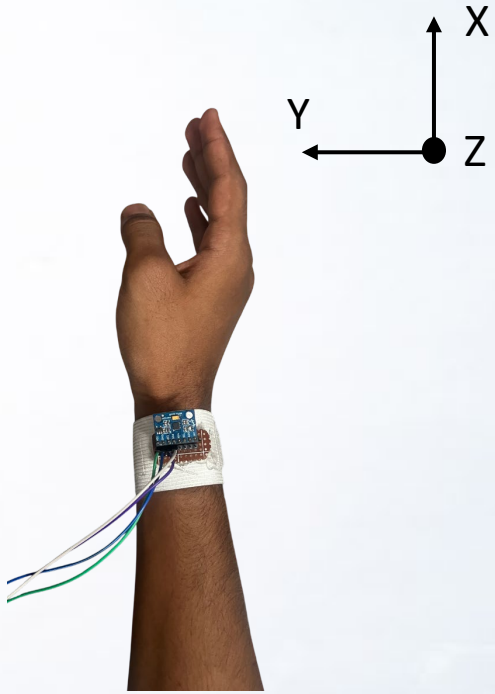


3. Algorithm Analysis

3.4 Wrist Control

The wrist is controlled using the roll angle measured from the accelerometer data mounted on the wrist.

$$Roll = \tan^{-1}\frac{a_y}{a_z} \times \frac{180}{\pi}$$



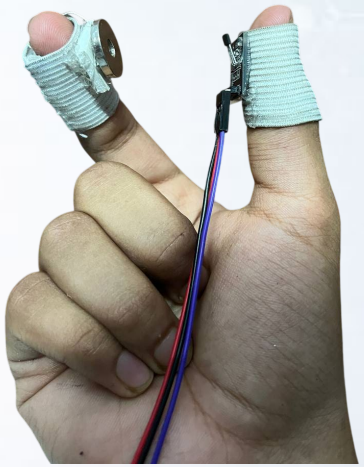
Flowchart :

3. Algorithm Analysis

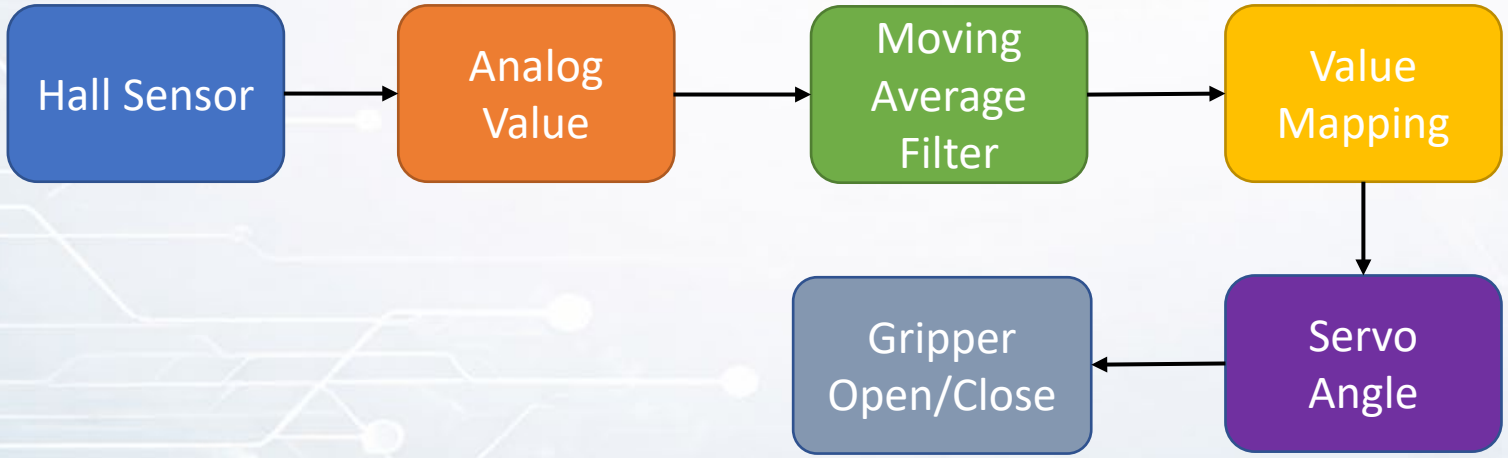
3.4 Gripper Control

The gripper opening is controlled by measuring magnetic distance using the Hall sensor and mapping that value to a servo angle

The analog data obtained at two extreme – i) Magnet farthest ii) Magnet closest is taken as the two limits of the angle to be mapped



Flowchart :



4. Code

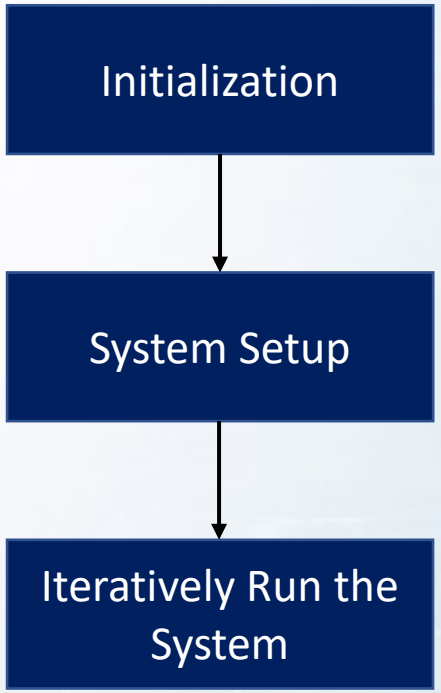
A single **Arduino Uno** is used to operate the whole system. The code follows the following architecture

Stage – 1: All library functions called and all the sensors initialized.

Stage – 2: All necessary variables defined and functions (Moving avg. filter, Compass heading calculation, etc.) needed at the start are defined.

Stage – 3: Inside "void setup", all MPU sensors and the compass sensor are initialized. The target heading for the compass sensor is calculated.

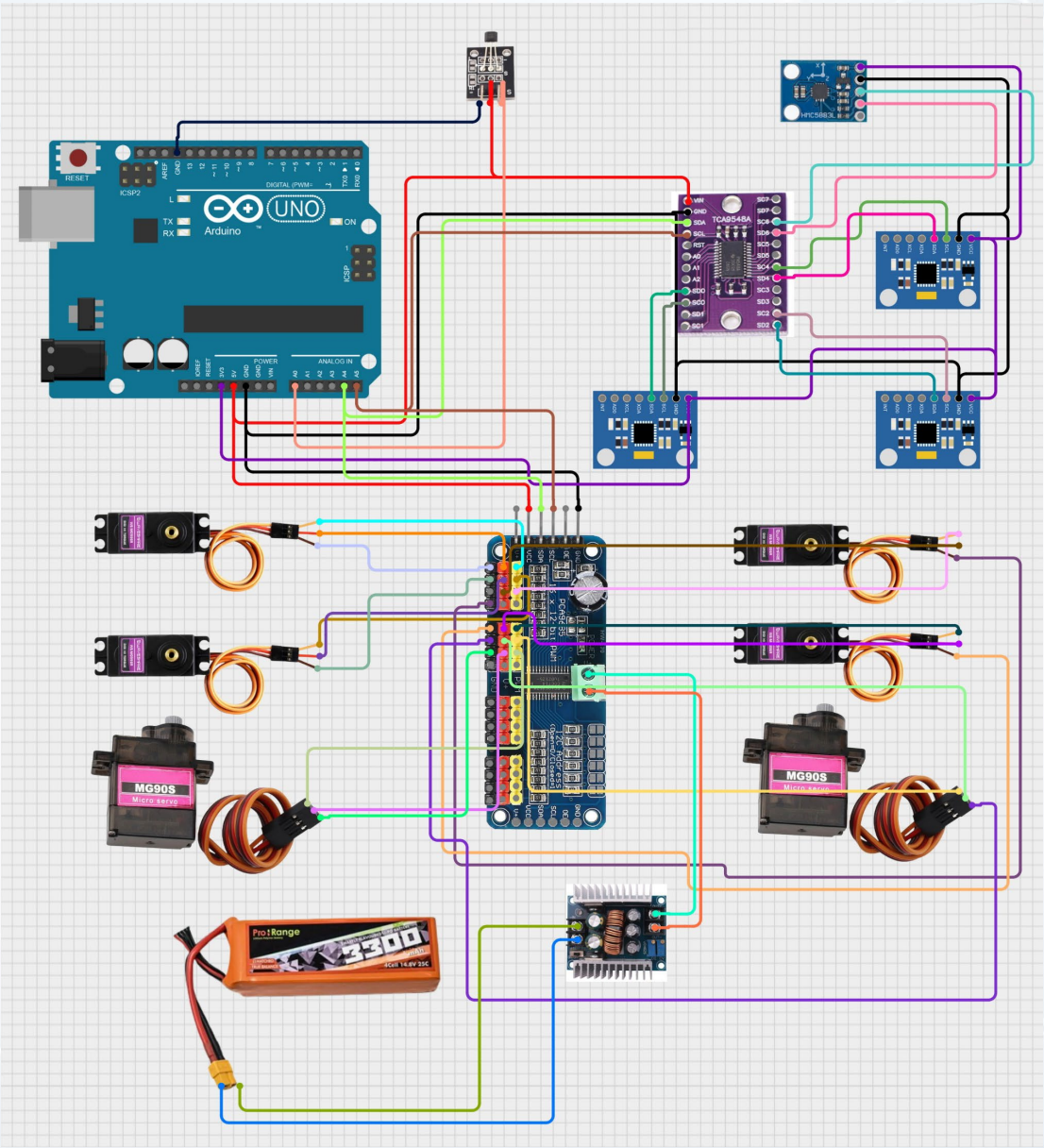
Stage – 4: In the "void loop" at first, the I2C mux is checked if it's working correctly, and then all the sensor data is processed one by one.



GitHub: https://github.com/intmushfiq/6-DOF-Robotic-Arm

4. CAD

4.1 Circuit Diagram



6. Cost

Mechanical & Structural	
3D Printed Body & Claw	4,247
Ball Bearings & Screw	650
Band & Glue	450
Screw driver & Cutter	500
Subtotal	**5,847 BDT**

Actuators & Drivers	
MG996R Servos (×4)	1440
SG90 Micro Servos (×2)	260
8-Ch PWM/Servo Driver	400
Arduino Uno	950
Subtotal	**3,050 BDT**

Sensors & Modules	
MPU6050 Accel/Gyro (×3)	610
GY-271 Compass Sensor	306
Hall Effect & Magnets	230
I2C Mux & TCA Module	250
Subtotal	**1,396 BDT**

Power , Connectivity & Others	
Buck Converters (20A+)	600
11.1V LiPo Battery	2100
Jumper Cables & Pins	280
Servo Extensions	240
Shipping and Others	400
Subtotal	**3,620 BDT**

TOTAL
13,913
BDT

PER PERSON
~2783
BDT

7. Environmental Concern

Low Energy Consumption
(Low-power sensors, motors and other components are used)

Reduced Human Exposure
(The system can help reduce direct human exposure to potentially hazardous environments)

Reduced Material Waste
(3D printed parts and custom-made arm band and sensor holders ensure reduced material usage)

Proper E-Waste Management
(Proper waste management protocol was followed)

8. Ethics

The 3D model of the robotic arm was sourced from an open source platform: https://github.com/bitluni/RobotArm/tree/master/Models

All other aspects of the project, including system design, circuit implementation, programming and testing were developed collaboratively by the team.

9. Reference

- Robot arm that mimics your arm movements. (n.d.). Retrieved from https://www.electromaker.io/project/view/robot-arm
- Bansode, puja. (2016). Robotic Arm Control through Human Arm Movement Using Accelerometers. International Journal of Advance Research and Innovative Ideas in Education.
- MEMS-Based IMU for pose Estimation. (n.d.). Retrieved from https://www.researchgate.net/publication/337695213_MEMS-Based_IMU_for_Pose_Estimation
- Islam, Md & Shajid-Ul-Mahmud, Md & Islam, Tariqul & Amin, Md & Haider, Mohammed Hossam. (2016). A low cost MEMS and complementary filter based attitude heading reference system (AHRS) for low speed aircraft. 1-5. 10.1109/CEEICT.2016.7873047.
- Wikipedia contributors. (2026, February 24). Heading (navigation). Retrieved from https://en.wikipedia.org/wiki/Heading_(navigation)

Thank You